Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Key Features:**
    - Customer demographics (Age, Gender, Location, Subscription Status)
    - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
    - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- **Missing Data:** 37 values in `Review Rating` column

3. Exploratory Data Analysis, Data Cleaning & Standardization using SQL

Data staging, profiling, and cleaning were handled directly within the database using structured SQL queries:

- **Staging Table Creation:** Created a dedicated analysis table (`csb`) duplicated from the raw dataset structure using `CREATE TABLE csb LIKE customer_shopping_behavior;` and populated it using a deep copy statement.

- **Column Standardization:** Restructured the tracking keys by modifying the primary identifier column. The corrupted source field name `ï»¿Customer ID` was altered and standardly renamed to `Customer_id`.

- **Duplicate Validation:** Validated dataset integrity by running a Common Table Expression (CTE) utilizing a window ranking function:

```
WITH cte AS (
    SELECT *, ROW_NUMBER() OVER(PARTITION BY Customer_id ORDER BY
Customer_id) AS row_num
    FROM csb
```

```sql
)
SELECT * FROM cte WHERE row_num > 1;
```

The check confirmed zero duplicate profiles within the collection.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_id	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	P M
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	14	Ve
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Ca
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	23	Cr
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	49	Pa
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	31	Pa
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	14	Ve
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	49	Ca
8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	19	Cr
9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	8	Ve
10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	4	Ca
11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	26	Ba
12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	10	Ba
13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes	37	Ve
14	65	Male	Dress	Clothing	51	New Hamp Delaware		Violet	Spring	4.7	Yes	Express	Yes	31	Pa
15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	34	De
16	64	Male	Skirt	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes	Store Pickup	Yes	8	Pa

- **Null & Blank Space Auditing:** Executed rigorous conditional checks across all 18 attributes combining `IS NULL` filters and text-trimming validations (`TRIM(column) = ''`). The data profile showed zero structural blanks or empty string entries.

- **Feature Engineering (Segmentation & Transformation):** * **Age Stratification:** Appended a categorical text attribute age_group and mapped individuals through conditional case routing thresholds:

  - `Age < 30` → `'yound_adult'`
  - `Age BETWEEN 30 AND 44` → `'adult'`
  - `Age BETWEEN 45 AND 64` → `'middle_age'`
  - `Remaining values` → `'seniour'`
  - **Frequency Numerical Mapping:** Generated an integer field `frequency_num` to assign standard operational time-step intervals against the textual `Frequency of Purchases` records (e.g., Weekly mapped to 7 days, Fortnightly/Bi-Weekly mapped to 14 days, etc.).
- **Redundancy Cleanup:** Safely dropped the unneeded `Promo Code Used` column from the staging database layout using an `ALTER TABLE` statement to preserve visual real estate for downstream visual analysis.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Amount	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases	age_group	frequency_num
	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	14	Venmo	Fortnightly	middle_age	14
	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Cash	Fortnightly	yound_adult	14
	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	23	Credit Card	Weekly	middle_age	7
	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	49	PayPal	Weekly	yound_adult	7
	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	31	PayPal	Annually	middle_age	30
	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	14	Venmo	Weekly	middle_age	7
	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	49	Cash	Quarterly	middle_age	90
	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	19	Credit Card	Weekly	yound_adult	7
	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	8	Venmo	Annually	yound_adult	30
	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	4	Cash	Quarterly	middle_age	90
	Arkansas	L	Purple	Fall	4.1	Yes	Store Pickup	Yes	26	Bank Transfer	Bi-Weekly	middle_age	14
	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes	10	Bank Transfer	Fortnightly	adult	14
	Delaware	M	Gold	Winter	4.5	Yes	Express Store Pickup		37	Venmo	Fortnightly	middle_age	14
	New Hampshire	M	Violet	Spring	4.7	Yes	Express	Yes	31	PayPal	Weekly	seniour	7
	New York	L	Teal	Winter	4.7	Yes	Free Shipping	Yes	34	Debit Card	Weekly	middle_age	7
	Rhode Island	M	Teal	Winter	2.8	Yes	Store Pickup	Yes	8	PayPal	Monthly	middle_age	365

4. Data Analysis using SQL (Business Transactions)

Structured relational analysis was executed to answer critical business metrics:

1. Revenue by Gender

Compared the total cumulative revenue generated between male and female customer demographics.

Gender	revenue
Male	155438
Female	75191

2. High-Spending Discount Users

Identified distinct customer profiles who leveraged active transaction discounts but still maintained a purchase metric strictly above the platform's overall average purchase amount

Result Grid | Filter Rows:

Customer_id	Purchase Amount (USD)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60

3. Top 5 Products by Rating

Isolated product lines that maintained the highest average customer review scores. | item_purchased

Item Purchased	avg_product_rating
Blouse	3.68
Sweater	3.76
Jeans	3.65
Sandals	3.84
Sneakers	3.76

4. Shipping Type Comparison

Evaluated differences in overall transaction size averages between standard delivery and express distribution channels.

Shipping Type	purchase
Express	60.34
Standard	58.41

5. Subscribers vs. Non-Subscribers

Analyzed total platform monetary support, user volume counts, and spend averages partitioned across distinct subscription statuses. | subscription_status | total_customers | avg_spend | total_revenue

Subscription Status	total_customer	avg_spend	total_spend
Yes	1033	59.42	61377
No	2830	59.81	169252

6. Discount-Dependent Products

Flagged the top five specific item categories that relied most heavily on promotional price reductions. | item_purchased | discount_rate

Item Purchased	discount_rate
Hat	49.67
Sneakers	49.66
Coat	48.43
Sweater	47.20
Pants	46.43

7. Customer Segmentation

Aggregated and bucketed standard users into New, Returning, and Loyal lifecycle tiers based on aggregate history metrics

customer_segment	number_of_customer
Loyal	3085
Returning	695
New	83

8. Top 3 Products per Category

Extracted the top three highest-velocity individual items bought inside each parent category.

item_rank	Category	Item Purchased	total_orders
1	Accessories	Jewelry	170
2	Accessories	Sunglasses	159
3	Accessories	Belt	159
1	Clothing	Blouse	169
2	Clothing	Shirt	168
3	Clothing	Pants	168
1	Footwear	Sandals	159
2	Footwear	Shoes	147
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	159

9. Repeat Buyers & Subscriptions

Assessed whether high-volume consumers exhibiting greater than 5 distinct purchases show higher correlation trends with standard subscription membership

Subscription Status	repeat_buyers
Yes	939
No	2503

10. Revenue by Age Group

Calculated total system sales generation broken down into standard age group divisions. | age_group.

age_group	total_revenue
middle_age	88406
adult	64807
yound_adult	52295
seniour	25121

5. Dashboard in Power BI

Finally, an interactive analytical reporting environment was engineered in Power BI utilizing the curated database structures to dynamically expose these behavioral dimensions.

The user interface reports core high-level metrics cleanly:

- **Total Customer Volume:** 3.9K unique records
- **Average Ticket Size:** $59.76 per transaction
- **Global Review Average:** 3.75 stars
- **Subscription Breakdown:** Visualizes a 27% Yes to 73% No split among consumers.

6. Business Recommendations

- **Boost Subscriptions:** Target the 73% non-subscriber base with targeted benefits to grow predictable, recurring subscription streams.
- **Customer Loyalty Programs:** Design personalized incentives for “Returning” users to convert them into high-value “Loyal” segments.
- **Review Discount Policy:** Optimize margin guardrails around highly discount-dependent product groups like hats and sneakers.
- **Product Positioning:** Showcase high-performing inventory lines across active media and store visual displays.
- **Targeted Marketing:** Focus acquisition and marketing budgets heavily toward high-revenue age brackets and express-shipping customer cohorts.